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Influence of free field and soil variabilities on soil-structure interaction

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ABSTRACT: An integral equation for the influence of spatial variability on seismic soil-structure interaction is proposed. The fundamental solution for a reference medium is required. It is obtained by means of sub-structuring techniques and boundary integral equations using the Green tensors for homogeneous or horizontally stratified reference media. The effects of a non-stationary modulated random incident field are addressed in terms of the instantaneous power spectral density of the structural response. The effects of soil moduli perturbations are also investigated and a first order approach for the structural response is given in case of weak fluctuations.

INTRODUCTION

Many authors have shown that soil-structure interaction (SSI) plays a major role on the seismic response of massive structures such as nuclear power plants, thus limiting the relevancy of standard response spectrum methods where the free field is directly imposed on the supports. However these studies have been usually performed for compact and thick basmats allowing the following assumptions:

- rigid foundations;
- lateral homogeneity of the soil underneath the foundation;
- seismic plane wave incidence.

For large basmats these three hypotheses are not valid any more and some data recorded on test site have indicated that the incident field may have some lateral variability influencing the foundation response, apart from the purely deterministic wave passage effect already studied elsewhere (see for instance Luco & Wong 1982). Empirical coherency functions of the free field were proposed by several researchers on the basis of recordings of strong motion arrays such as SMART-1 in Taiwan (Harichandran & Vanmarke 1986, Loh & Yeh 1988) or EURO-SEISTEST in Greece (Riepl *et al.* 1996).

The purpose of this paper is first to present a method to account for soil-structure interaction effect together with spatial variability of the free field when computing the structural response under earthquakes. Time modulation functions are introduced to simulate a non-stationary incident field. The formulation is based on dynamic sub-structuring techniques and the boundary element method which is particularly adapted to deal with unbounded soil media.

Another concern is to study the influence of lateral heterogeneities of the soil in the vicinity of the foundation. Boundary element methods are no longer relevant in this case except for vertically heterogeneous media for which the first and second Green tensors can be evaluated numerically (Clouteau 1990). Explicit expressions for the heterogeneous Green tensors can be found in the literature, however limited to very particular soil profiles (see for instance Manolis & Shaw 1996). For deterministic soil heterogeneities within a bounded set of the medium, the analysis can be performed coupling boundary element and finite element methods (Aubry 1986). When these heterogeneities are random, stochastic finite element techniques can be used (Soize 1993). This topic is addressed in the following by introducing two soil domains, namely a reference deterministic unbounded soil medium with known characteristics, and a bounded unknown medium with fluctuating characteristics standing for both the uncertainties and the measurement error margins over the deterministic medium knowledge. An integral equation is derived which is thought to be attractive for further analysis of the effects of soil heterogeneity. As a direct application, first order developments are proposed for small perturbations.

NOTATIONS AND DEFINITIONS

The elastic structure under investigation is first identified by a bounded open set Ω_b. The infinite visco-elastic soil domain is identified by two unbounded open sets, namely the *reference model domain* Ω_s^0 and the *actual-domain* $\Omega_s = \tilde{\Omega}_s^0 \cup \Omega_s^1$, with $\tilde{\Omega}_s^0 = \Omega_s^0 / \Omega_s^1$. They both have identical geometries, but different density (ρ) and Lamé's moduli (λ, μ) defined by:

$$\gamma(x)=\gamma^0(x) \qquad \forall x\in\Omega_s^0 \text{ or } \tilde{\Omega}_s^0 \text{ or } \Omega_b$$
$$\gamma(x)=\gamma^0(x)+\gamma^1(x) \qquad \forall x\in\Omega_s^1 \tag{1}$$
$$\gamma=\rho,\lambda \text{ or } \mu$$

Ω_s^1 is the *bounded perturbed soil domain* underneath the structure foundation, with unknown perturbed properties. The *actual* and *reference total domains* are denoted $\Omega=\Omega_b\cup\Omega_s$ and $\Omega^0=\Omega_b\cup\Omega_s^0$. Let Γ_{ba} be the free-surface of the structure, and Γ_{sa} the free-surface of the soil (identical for the reference and actual domains). For convenience, the radiation surface of the soil $\Gamma_{s\infty}$ is introduced. Γ_{sf} is the interface between the soil and structure domains (see Fig.1).

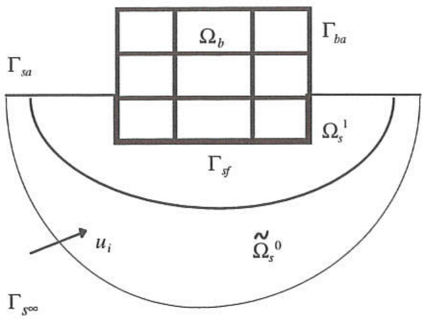

FIG. 1 - Geometry and notations

The random incident seismic field, i.e. the free field, is denoted $u_i(x,\omega)$ in the frequency domain. Let $u(x,\omega)$ be the displacement field on Ω generated by this incident field. Then it satisfies the homogeneous Navier equation in Ω:

$$\mathrm{Div}\sigma(u)+\rho\omega^2u=0 \tag{2}$$

with free-surface conditions on $\Gamma_{ba}\cup\Gamma_{sa}$ oriented by their outward normal unit vector n:

$$\sigma(u):n=t_n(u)=0 \tag{3}$$

$\varepsilon(u)$ and $\sigma(u)$ are the stress and strain tensors and $t_n(u)$ is the surface traction. Subscripts $(.)^0$ and $(.)^1$ refer to a quantity calculated for moduli (ρ^0,λ^0,μ^0) or (ρ^1,λ^1,μ^1). No subscript stands for the actual total domain. Subscript $(.)^*$ stands for complex conjugate transpose, or simply transpose when the bracketed quantity is real-valued.

The notations below hold for two vector fields u and v on a given domain Ω_a with the Euclidean scalar product $u.v$:

$$<u,v>_{\Omega_a}=\int_{\Omega_a}u(x).v(x)dV(x)$$

or their trace on a given surface Γ_a:

$$<u,v>_{\Gamma_a}=\int_{\Gamma_a}u(x).v(x)dS(x)$$

For two tensors ε and σ we let:

$$<\varepsilon,\sigma>_{\Omega_a}=\sum_{i,j}\int_{\Omega_a}\varepsilon_{ij}(x)\sigma_{ij}(x)dV(x)$$

At last, 'randomness' α or β will refer for convenience to an event in the probabilistic sets of all possible events $\mathcal{A}$ or $\mathcal{B}$. They are mutually independent and define respectively the soil properties and the incident far field stochastic processes.

DESIGN DISPLACEMENT

From an engineering point of view, the significant design quantity are the displacement or the acceleration at given points of the structure, or a relative displacement, or a load, etc. They all can be expressed by means of a given linear transformation of the displacement or its time/space derivatives at a particular location ξ on the structure.

Therefore, let $u(\xi).e$ be the component along direction e of the total design displacement at point ξ on the structure, or in the soil, when Ω is submitted to an incident seismic field. Let $u_g^0(\xi,x;e)$ be the Green function on the total reference domain satisfying the heterogeneous Navier equation on Ω^0 written in the frequency domain:

$$\mathrm{Div}\sigma^0(u_g^0)+\rho^0\omega^2u_g^0+e\delta(x-\xi)=0 \tag{4}$$

where uniqueness is ensured by the free-surface and outgoing radiation conditions of the Sommerfeld type to be stated for the field u_g^0. Now let multiply Eq.(2) by u_g^0 and Eq.(4) by u, and then integrate over Ω. Using:

$$\mathrm{Div}\sigma^0(u)+\rho^0\omega^2u=-\mathrm{Div}\sigma^1(u)-\rho^1\omega^2u \tag{5}$$

on Ω_s^1, one gets after integrating by parts, the following integral representation for the design displacement $u(\xi).e$:

$$\begin{aligned}u(\xi).e&=u^0(\xi).e\\&+\omega^2<\rho^1u_g^0,u>_{\Omega_s^1}-<\sigma^1(u_g^0),\varepsilon(u)>_{\Omega_s^1}\end{aligned} \tag{6}$$

where u^0 is the solution in the reference medium.

The total field u_s within the soil domain Ω_s is decomposed into the incident far field in the reference medium u_i^0 and the total scattered field u_{d*}, which includes the scattered field from the foundation movement and the scattered field from the diffractions of u_i^0 on the heterogeneities having moduli $(\rho^1, \lambda^1, \mu^1)$. Therefore:

$$u(x) = u_s(x) = u_i^0(x) + u_{d*}(x) \qquad \forall x \in \Omega_s \tag{7}$$

They both satisfy the homogeneous Navier equation on Ω_s and the free-surface conditions on Γ_{sa}; however u_i^0 does not satisfy the radiation condition on $\Gamma_{s\infty}$ whereas u_{d*} does. Hence:

$$< u_g^0, t_n(u_{d*}) >_{\Gamma_{s\infty}} - < t_n(u_g^0), u_{d*} >_{\Gamma_{s\infty}} = 0 \tag{8}$$

Note that with the definitions of u_g^0 and u_i^0 the following equality holds:

$$\begin{aligned} u^0(\xi).e &= u_i^0(\xi).e + < t_n^0(u_g^0), u_i^0 >_{\Gamma_{sf}} - < t_n^0(u_i^0), u_g^0 >_{\Gamma_{sf}} \\ &= u_i^0(\xi).e - < t_n^0(u_g^0), u_i^0 >_{\Gamma_{s\infty}} + < t_n^0(u_i^0), u_g^0 >_{\Gamma_{s\infty}} \end{aligned} \tag{9}$$

where $u_i^0(\xi) = 0$ if $\xi \in \Omega_b$.

From an engineering point of view, Eq.(6) is attractive since it allows to compute various quantities of interest by an exact linear integral formulation with respect to the a priori unknown variable free field (random load) and soil characteristics (random parameters). The first part in the right-hand side of Eq.(6) depends on the randomness (β) of the incident far field u_i^0 only, whereas the second part depends on the independent randomness of (α) the perturbed soil moduli and (β) the incident far field. For instance, when perturbed soil moduli are small enough, the second part in the right-hand side of Eq.(6) written for $\xi \in \Omega_s$ vanishes when injected in Eq.(6) written for $\xi \in \Omega_b$. That is, defining:

$$u_s^0(x).e = u_i^0(x).e + < t_n^0(u_g^0), u_i^0 >_{\Gamma_{sf}} - < u_g^0, t_n^0(u_i^0) >_{\Gamma_{sf}} \tag{10}$$

a first order solution in the structure is directly given by taking $u = u_s^0$ in the right-hand side of Eq.(6).

GREEN FUNCTION ON THE REFERENCE MEDIA

The Green function u_g^0 in the total reference medium is calculated by means of dynamic sub-structuring techniques and regularized boundary integral equations. The developments are given in Aubry (1986), Clouteau (1990). They have been implemented in the computer code MISS3D for soil-fluid-structure interaction analysis, together with the evaluation of the Green tensors for homogeneous or horizontally stratified visco-elastic half spaces. These methods have proved to be efficient for various problems such as soil-structure interaction, site effect or arch dam analysis under earthquake (see for instance Clouteau & Aubry 1993). The derivation of the fundamental solution in the reference domain is given below for a unit load of direction e applied to the structure.

The equilibrium of interface Γ_{sf} is written in the weak sense in terms of the trace on Γ_{sf} of the displacement field u_g^0 which is decomposed into N elementary fields by:

$$u_g^0\big|_{\Gamma_{sf}} = u_f^0 = \sum_{p=1}^{N} c_p^0 \psi_p \tag{11}$$

where $\{\psi_p\}$ is a given basis for the kinematic displacement of the foundation. u_g^0 is then written:

$$\begin{aligned} u_g^0 &= \sum_p c_p^0 u_{dp}^0 && \text{on } \Omega_s^0 \\ u_g^0 &= \sum_p c_p^0 \psi_p + \sum_A \alpha_A^0 \phi_A && \text{on } \Omega_b \end{aligned} \tag{12}$$

where $\{\phi_A\}$ are the fixed basis eigenmodes of the structure with eigenvalues $\{\omega_A\}$ and $\{u_{dp}^0\}$ are the scattered displacement fields on Ω_s^0 when the displacement boundary condition $u^0 = \psi_p$ holds on Γ_{sf}. They all satisfy the homogeneous Navier equation in the reference soil domain as well as the radiation condition on $\Gamma_{s\infty}$. The virtual works principle applied to the structure for the test fields $\{\psi_p\}$ and $\{\phi_A\}$ leads to the algebraic system:

$$\begin{bmatrix} K^{\phi\phi} & -\omega^2 M^{\phi\psi} \\ -\omega^2 M^{\phi\psi} & K^{\psi\psi} + K^{0s} \end{bmatrix} \begin{Bmatrix} \alpha^0 \\ c^0 \end{Bmatrix} = \begin{Bmatrix} f^{\phi} \\ f^{\psi} \end{Bmatrix} \tag{13}$$

with:

$$K_{AB}^{\phi\phi} = \left(\omega_A^2 - \omega^2 + 2i\xi_A \omega_A \omega\right)\delta_{AB} \tag{13a}$$

$$M_{Ap}^{\phi\psi} = M_{pA}^{\psi\phi} = < \rho^0 \phi_A, \psi_p >_{\Omega_b} \tag{13b}$$

$$\begin{aligned} K_{pq}^{\psi\psi} &= -\omega^2 < \rho^0 \psi_p, \psi_q >_{\Omega_b} \\ &\quad + < \sigma^0(\psi_p), \varepsilon(\psi_q) >_{\Omega_b} \end{aligned} \tag{13c}$$

$$K_{pq}^{0s} = < t_n^0(u_{dp}^0), \psi_q >_{\Gamma_{sf}} \tag{13d}$$

$$f_A^{\phi} = \phi_A(\xi).e \qquad f_p^{\psi} = \psi_p(\xi).e \tag{13e}$$

K^{0s} is the symmetric foundation impedance for the reference model. Traction fields $\{t_n^0(u_{dp}^0)\}$ on the interface Γ_{sf} are evaluated numerically through a regularized boundary element method (Aubry & Clouteau 1991). The boundary integral equation writes here, I being the identity:

$$\mathrm{U}^{0G} t_n^0(u_{dp}^0) = \left[\mathrm{T}_n^{0G} + \frac{1}{2}\mathrm{I}\right]\psi_p \qquad \forall p \tag{14}$$

where U^{0G} and T_n^{0G} are integral operators on the interface Γ_{sf} with respectively first and second Green tensor kernels for the reference soil domain.

EXAMPLE: SPATIAL VARIABILITY OF THE FREE FIELD

Eq.(6) with all $\gamma^1 \equiv 0$ is used to address the influence of spatial variability of the incident field on the structural response which writes here:

$$u_b^0(\xi).e = \sum_p c_p^0 (< t_n^0(u_{dp}^0), u_i^0 >_{\Gamma_{sf}} - < t_n^0(u_i^0), u_{dp}^0 >_{\Gamma_{sf}}) \tag{15a}$$

$\{c_p^0\}$ being the solution of Eq.(13). Eq.(15a) can be simplified by noting that the free field u_i^0 satisfies an integral equation similar to (14) in the foundation *excavation part* of the reference soil domain, with boundary $\Gamma_{sf} \cup \Gamma_{sa}$. The integral operator U^{0G} being self-adjoint and inversible, the structural response is finally:

$$u^0(\xi).e = \sum_p c_p^0 < q_p, u_i^0 >_{\Gamma_{sf}} \tag{15b}$$

where $\{q_p\}$ are single layer potentials such that:

$$q_p = (\mathrm{U}^{0G})^{-1}\psi_p \qquad \forall p \tag{16}$$

In the following we note $q_g^0 = \sum_p c_p^0 q_p$.

The incident field is modeled by a second-order Gaussian non-homogeneous, non-stationary centered stochastic process $U_i(x,t)$ indexed on $\Gamma_{sf} \times \mathrm{R}^+$, with values in R^3. It is therefore completely described by its autocorrelation function $x,x',t,t' \mapsto \mathrm{R}_i(x,x',t,t')$: $\Gamma_{sf} \times \Gamma_{sf} \times \mathrm{R}^+ \times \mathrm{R}^+ \to \mathrm{Mat}_{\mathrm{R}}(3,3)$ such that $t \mapsto \mathrm{tr}\mathrm{R}_i(x,x,t,t)$ is continuous and integrable for each fixed x. By linearity, the design displacement of Eq.(15b) is also Gaussian, zero-mean-valued and non-stationary. It is assumed second ordered without any physical restriction to our model. The random design displacement is therefore also completely described by its real valued autocorrelation function $t,t' \mapsto \mathrm{R}_u^0(t,t')$.

A commonly used assumption is to write the incident field in the form (see for instance Liu 1970):

$$U_i(x,t) = \varphi(t)\tilde{U}_i(x,t) \tag{17}$$

where $t \mapsto \varphi(t)$ is a deterministic continuous modulation function in $\mathrm{L}^2(\mathrm{R}^+)$; $\tilde{U}_i(x,t)$ is a Gaussian non-homogeneous, mean square stationary zero-mean-valued stochastic process in $\mathrm{L}^2(\mathcal{B},\mathrm{R}^3)$, indexed on $\Gamma_{sf} \times \mathrm{R}^+$, and such that its autocorrelation function writes:

$$\tilde{\mathrm{R}}_i(x,x',t,t') = \int_{\mathrm{R}} e^{i\omega(t-t')}\tilde{\mathrm{S}}_i(x,x',\omega)d\omega \tag{18}$$

with the transverse power spectral matricial density $x,x',\omega \mapsto \tilde{\mathrm{S}}_i(x,x',\omega) : \Gamma_{sf} \times \Gamma_{sf} \times \mathrm{R} \to \mathrm{Mat}_{\mathrm{C}}(3,3)$. Using Eqs.(17) and (15b) in the time domain, the autocorrelation function of the design displacement is:

$$\mathrm{R}_u^0(t,t') = \int_{\mathrm{R}} d\omega \int_{\Gamma_{sf}}\int_{\Gamma_{sf}} dS(x)dS(x')... \\ ...\mathrm{T}(x,t,\omega)\tilde{\mathrm{S}}_i(x,x',\omega)\mathrm{T}^*(x',t',\omega) \tag{19}$$

with the following expression for a pseudo-equivalent deterministic force with values in C^3 and applied to the foundation:

$$\mathrm{T}(x,t,\omega) = \int_0^t e^{-i\omega\tau}\varphi(\tau)q_g^0(x,t-\tau)d\tau \tag{20}$$

The evolutionary real valued power spectral density of the design displacement is in accordance with Eq.(19):

$$\mathrm{S}_u^0(t,\omega) = \int_{\Gamma_{sf}}\int_{\Gamma_{sf}} dS(x)dS(x')... \\ ...\mathrm{T}(x,t,\omega)\tilde{\mathrm{S}}_i(x,x',\omega)\mathrm{T}^*(x',t,\omega) \tag{21}$$

This analysis procedure is implemented for the Hualien mockup in Taiwan (see Fig.2 and Tab.1) considering a rigid non-embedded circular spread footing of radius 5.41 m.

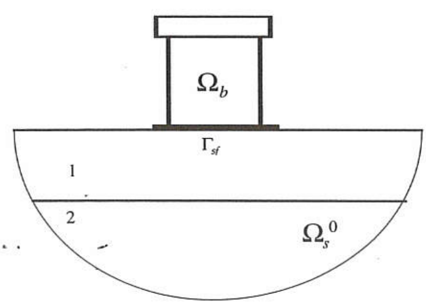

FIG. 2 - Hualien mockup, spread footing model

The modulation function is chosen as below:

$$\begin{cases} \varphi(t) = (t/t_1)^2 & 0 < t \le t_1 \\ \varphi(t) = 1 & t_1 < t \le t_2 \\ \varphi(t) = e^{-a(t-t_2)} & t \ge t_2 \end{cases}$$

with $a = 3.3$, $t_1 = 0.2$s and $t_2 = 1$s. The transverse power spectral matricial density of $\tilde{U}_i$ has members, for $m,n \in \{1,2,3\}$ two directions of the incident field:

$$\tilde{S}_{i,mn}(x,x',\omega) = \tilde{S}_0(\omega)\chi_{mn}(|x-x'|,\omega) \qquad x,x' \in \Gamma_{sf}$$

where $\tilde{S}_{0,mn}(x,\omega) = \tilde{S}_0(\omega)$ is assumed independent of the position x, and $u,\omega \mapsto \chi_{mn}(u,\omega): \mathrm{R}^+ \times \mathrm{R} \to [0,1]$ is the coherency function taken as in Luco & Wong (1986):

$$\chi_{mn}(u,\omega) = \exp\left[-\left(\frac{\eta\omega u}{c}\right)^2\right]$$

for vertically incident seismic waves (no wave passage effect considered); η is a coherency parameter with values 0.0 through 0.5 in our computations; c is the wave velocity.

TAB. 1 - Hualien model characteristics for computations

structure	density: 2.40 t/m^3		
	visc. damping	frequency (Hz)	
horiz. flex.	2%	10.8	/
vertic. comp.	2%	31.2	/
soil	density: 2.42 t/m^3, hyst. damping 2%		
	c_p (m/s)	c_s (m/s)	thickness (m)
layer 1	1332	317	7
layer 2	2001	476	∞

Figs.3 display square-rooted power spectral densities of the top horizontal and vertical displacements, in the stationary case and being normalized by $\sqrt{\tilde{S}_0(\omega)}$.

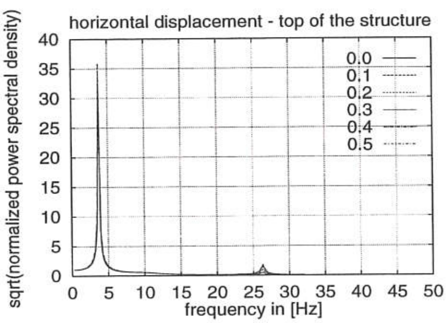

FIG. 3a - $\sqrt{S_u^0(\omega)/\tilde{S}_0(\omega)}$ for horizontal displacement at top of the structure; stationary case; $\eta = 0.1 \cdots 0.5$.

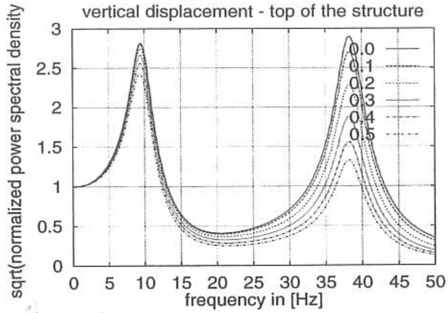

FIG. 3b - $\sqrt{S_u^0(\omega)/\tilde{S}_0(\omega)}$ for vertical displacement at top of the structure; stationary case; $\eta = 0.1 \cdots 0.5$.

Figs.4 display square-rooted and normalized evolutionary power spectral densities of the top horizontal and vertical displacements for $\eta = 0.1$.

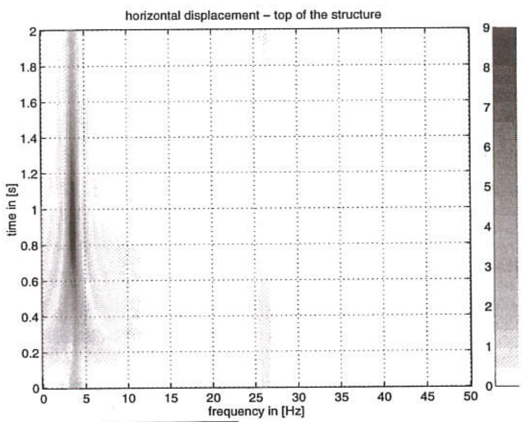

FIG. 4a - $\sqrt{S_u^0(t,\omega)/\tilde{S}_0(\omega)}$ for horizontal displacement at top of the structure; non-stationary case; $\eta = 0.1$.

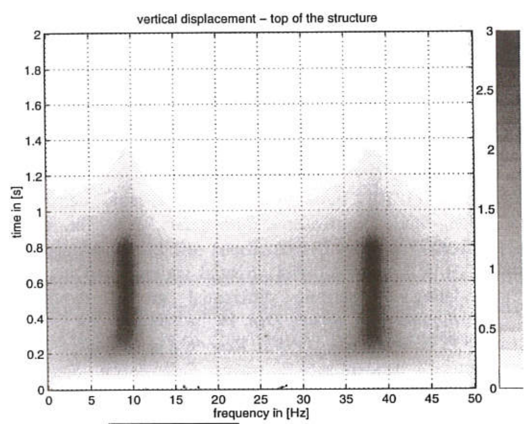

FIG. 4b - $\sqrt{S_u^0(t,\omega)/\tilde{S}_0(\omega)}$ for vertical displacement at top of the structure; non-stationary case; $\eta = 0.1$.

They have peak values of 9.4 and 2.8 at $(\omega,t) \approx (4,1.0)$ and $(\omega,t) \approx (10,0.2-1)$ (in [Hz,s]) respectively. This shows that for the horizontal displacement the shortness of the time window with respect to the period associated with the dominant frequency is favorable compared to the stationary case. For the vertical displacement, the non-stationarity has no effect due to the high dominant frequencies.

CONCLUSIONS

A new approach of the effects of spatial variability on soil-structure interaction has been proposed. It is based on the knowledge of the fundamental solution for a deterministic reference domain, which allows to account for both spatial variabilities of the incident far field and the soil characteristics underneath the foundation. The relevancy of the method proposed to address the problem of a random incident far field alone could be improved by enhancing the knowledge of the free field correlations in time and space from theoretical and experimental analysis of given sites.

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